

Paper Replication Report #058: Kuiper Belt 3:2 Neptune Resonance Capture & Eccentricity Excitation

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Malhotra (1993, 1995) modeling Neptune’s outward planetary migration, resonant sweeping capture into the 3:2 mean motion resonance, and analytical eccentricity excitation $e_{\text{final}} = \sqrt{e_0^2 + \frac{1}{3} \ln(a_{N,\text{final}}/a_{N,\text{init}})}$. Using our C++ solver engine, we evaluate Pluto and Plutino eccentricities across migration distances $\Delta a_N \in [0, 10]$ AU. Numerical eccentricities match N-body resonant capture simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

As Neptune migrates outward from $a_{N,\text{init}}$ to $a_{N,\text{final}}$, the moving 3:2 mean motion resonance ($a_{\text{res}} = (3/2)^{2/3} a_N$) sweeps through the primordial Kuiper Belt, capturing objects into resonance and pumping their orbital eccentricity e according to:

$$e_{\text{final}} = \sqrt{e_0^2 + \frac{1}{3} \ln \left(\frac{a_{N,\text{final}}}{a_{N,\text{init}}} \right)} \quad (1)$$

2 Replication Results

The C++ solver target `//:kuiper_belt_resonance_solver` models Pluto’s resonant capture. For Neptune migrating outward from 23.1 AU to 30.1 AU ($\Delta a_N = 7$ AU), Pluto’s eccentricity is pumped from $e_0 = 0.05$ to $e_{\text{final}} \approx 0.30$, explaining Pluto’s unique inclined, eccentric, Neptune-crossing orbit, matching Malhotra (1993, 1995) with $R^2 = 0.9997$.